
PO12Q - Quantitative Political Analysis: Uncovering Relationships

Week 1 - Worksheet Solutions

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1.
 - a. **Temporal precedence:** If x occurs before y , then y cannot cause x .
 - b. **Confounding:** A variable x_2 that creates a spurious association between x_1 and y .
 - c. **Intervening:** An intervening variable is a variable that explains the relationship between two other variables, such that x affects m , and in turn, m affects y .
 - d. **Interaction (or moderator):** An interaction effect exists when the association between one predictor, x , and an outcome, y , is conditional on another predictor, z .
 - e. **Spurious correlation (or common cause):** A relationship between two variables, x_1 and y , is said to be spurious when it appears to exist in a simple analysis but disappears or weakens substantially once a third variable, x_2 , is introduced into the analysis.
 - f. **Manipulation:** If a variable x is manipulated in an experimental design, then it must be the cause.
 - g. **Mechanism:** The mechanisms approach focuses on identifying the processes or pathways through which cause and effect are linked.
2.
 - a. **False.** Statistical significance indicates an association, not causality; causal inference also requires asymmetry, theoretical relevance, and the exclusion of alternative explanations.
 - b. **True.** In an intervening variable relationship, Z is part of the causal mechanism linking X to Y and should be explained rather than controlled away. c. **True.** Asymmetry rules out reverse causation within the same theoretical explanation, requiring that causes precede effects.
 - c. **True.** Multiple explanations imply that x_1 and x_2 each have independent causal effects on Y , and they need not be correlated with each other.
 - d. **True.** The counterfactual approach evaluates causality by considering what would have happened to the outcome if the cause had not occurred.
 - e. **True.** An interaction effect exists when the effect of x_1 on y varies depending on the value of a third variable x_2 .
3.
 - a. Asymmetry → iii. The cause must precede the effect and not vice versa
 - b. Alternative Explanations → iv. The process of ruling out spurious or confounding variables
 - c. Mechanism → i. The process or “black box” through which x produces y
 - d. Empirical Context → The specific conditions or environment where the cause occurs

4.

a. Spurious Relationship:

- z is a common cause of both x and y .
- Arrows: $z \rightarrow x, z \rightarrow y$.
- Remove the direct arrow between x and y (no direct causal effect).

b. Intervening Variable (Mechanism):

- x causes y indirectly through z .
- Arrows: $x \rightarrow z \rightarrow y$.
- Remove the direct arrow from x to y .

c. Multiple Explanations:

- Both x and z independently cause y .
- Arrows: $x \rightarrow y, z \rightarrow y$.
- There may or may not be an arrow between x and z ; the arrow is required if x and z are interdependent.

d. Interaction Effect:

- The effect of x on y depends on the level of z .
- Arrows: $x \rightarrow y, z \rightarrow$ arrow between x and y .

e. Spurious Association:

- z is the common cause making the correlation between x and y misleading.

f. Intervening variable:

- x is the primary cause, but its effect is transmitted through z rather than operating directly on y .

5. (Essay Guidance):

A researcher might prefer the “Manipulation” path to causality, because it involves actively intervening in the system – such as implementing a policy change in a randomized trial or pilot program. Manipulation allows the researcher to observe the direct effect of the intervention on the outcome, providing stronger evidence that the policy *produced* the change and helping to rule out spurious correlations.

In contrast, the “Neo-Humean Regularity” approach relies on observing that the policy and outcome consistently co-occur across multiple instances or repeated observations. While regularity can suggest a pattern, it cannot on its own establish that the policy caused the outcome, since the correlation might be due to confounding factors or chance. Manipulation can demonstrate causality even with a single well-designed experiment, whereas regularity requires many repeated cases to infer a consistent pattern.

6. To move from a statistical association between the number of police officers and crime rates to a credible causal explanation, the researcher must address all elements of the causal framework:
 - a. **Relevance within theoretical and empirical context:**
 - i. **Empirical Context:** Specify the parameters within which the causal claim is to be made, such as the specific city, its structure and composition, the time-frame, etc.
 - ii. **Clear Theoretical Framework:** Explain why more police should reduce crime (e.g., a mechanism such as deterrence, or rapid response).
 - iii. **Clear Conceptualization:** Define precisely what is meant by “more police” (number of officers, per capita, deployment patterns) and “crime” (types, measurement).
 - iv. **Exclusion of Alternative Explanations:** Control for confounding factors such as poverty, economic shifts, policing policies, or demographic changes.
 - b. **Asymmetry:** Demonstrate that increasing police reduces crime, rather than high-crime areas simply attracting more police (reverse causality), or both happening at the same time (simultaneity).
 - c. **Significant Statistical Association:** Establish that the effect is not only statistically significant but substantively strong, and robust across different model specifications.
7.
 - a. The claim fails to sufficiently satisfy several criteria for causality. In particular, it does not adequately exclude alternative explanations, does not clearly establish asymmetry, and provides only a weakly specified mechanism.
 - b. The primary concern is the exclusion of alternative explanations. Neighbourhoods with more street lighting may differ systematically from others in terms of income, policing levels, community organisation, or prior crime trends.
 - c. The causal claim would be strengthened by evidence from a research design that approximates manipulation, such as a randomized rollout of street lighting. The randomisation would help ensure that the treatment and control groups are comparable, thereby reducing the risk of confounding factors influencing the observed relationship between street lighting and crime rates. The manipulation itself introduces agency into the causal claim, allowing for a more credible inference that the presence of street lighting directly affects crime rates, rather than being correlated due to other underlying factors.
8.
 - An indirect causal structure would involve unemployment increasing crime through mechanisms such as financial strain or reduced opportunity costs of offending. This can be represented as $x \rightarrow z \rightarrow y$, where unemployment (x) first leads to financial strain (z), and then to crime (y).
 - A spurious relationship could arise if a third variable, such as economic decline or deindustrialization (z), causes both higher unemployment and higher crime. In this case, the appropriate structure is $z \rightarrow x$ and $z \rightarrow y$, with no direct causal arrow between unemployment and crime.
 - An interaction structure would involve the effect of unemployment on crime depending on a third variable, such as social welfare provision or community cohesion (z). Here, unemployment affects crime, but the strength or direction of this effect varies by the level of z .

9.

- The first statement represents a valid counterfactual causal claim: “If the policy had not been implemented, the observed improvement in outcomes would not have occurred.” This statement directly compares the observed outcome to a hypothetical scenario in which the cause is absent.
- The second statement does not support a causal interpretation because it reverses the direction of causality. It suggests that changes in outcomes caused the policy to be adopted, rather than the policy causing the change in outcomes.
- The third statement does not establish causality because it introduces plausible alternative explanations without specifying how the effect of the policy is isolated from these other influences. In practice, this may be a reasonable approach to take, but it does not constitute a counterfactual claim that attributes the outcome to the policy.